



Natural compensation mechanism of soil water infiltration through decayed roots in semi-arid vegetation species

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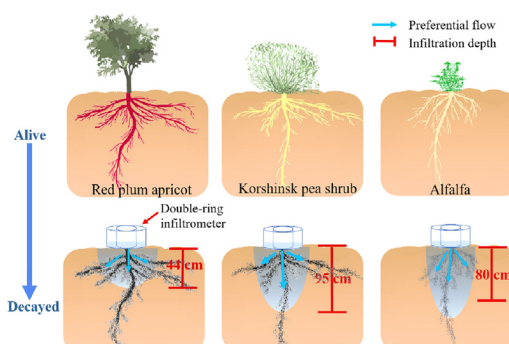
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HIGHLIGHTS

- Decayed plant root channels significantly increased semi-arid soil infiltration capacity.
- Preferential water flow by decayed root channels potentially relieves soil dry layer.
- Decayed root channels appear as a natural compensation measure of dried soil layers.

GRAPHICAL ABSTRACT



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ABSTRACT

In arid and semi-arid areas, unsuitable afforestation programs may cause soil water deficit and even, in some cases, deep-soil desiccation and the subsequent death of part of the vegetation. However, the benefits of preferential flow through decayed root channels for supplying water in the dried soil layers are rarely evaluated. This study examines the effects of root channels on soil water content (SWC) and infiltration (initial and steady-state) in two planted species with high drought tolerance, namely: Korshinsk peashrub (*Caragana korshinskii* Kom.), and red plum apricot (*Armeniaca vulgaris* Lam.), using bare land as control site and published data from alfalfa (*Medicago sativa* L.) plots to refine results analysis. The infiltration rates were measured using a double-ring infiltrometer in areas with alive and decayed roots; and methylene blue was used to trace the pathways of water flow. The highest SWC appeared in the Korshinsk peashrub land, whereas the alfalfa grassland had the lowest SWC; and the differences of SWC among treatments were significant at all soil layers. Korshinsk peashrub had the highest steady infiltration rates, which were about 23% and 83% higher than those rates measured in the fruit tree plantation and alfalfa grasslands, respectively. The steady infiltration rates were significantly and positively correlated with the average root diameter and area. Within the same species, the decayed root plots significantly increased the soil water infiltration capacity compared to the alive root plots. Considering the water scarcity conditions of the study area, the preferential flow formed through decomposed roots was conducive to soil water supplement,

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appearing as a natural compensation mechanism caused by dead vegetation. Our findings contribute to a better understanding of soil water infiltration during root decay of the plants in water-limited ecosystems, and how this dynamic replenishes soil moisture and alleviates soil desiccation.

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1. Introduction

Soil water scarcity is the key limiting factor for vegetation sustainability in drylands, including natural vegetation and afforestation. However, it is expected that droughts are going to be more severe in drylands over the coming years owing to climate change (Brown, 2002; Breshears et al., 2005). Recent studies have demonstrated that intensive afforestation activities in semi-arid regions that are in excess of local water availability have caused imbalances in the soil water budget (e.g. Jia et al., 2017). High density stands lead to soil water deficit and even favour dried soil layers (DSL), which cause the death of part of the vegetation in arid and semi-arid areas (Jia et al., 2017; Huang et al., 2018). DSL are mainly caused by chronic shortage of rainwater, high evapotranspiration and excessive consumption of deep soil water by plants (Jipp et al., 1998; Huang et al., 2018). DSL have a negative impact on the water cycle in the soil–plant–atmosphere continuum by restraining water interchanges between the upper and deeper soil layers (Lucero et al., 2000; Valdez-Hernandez et al., 2010). Besides, the blocked layer caused by DSL may affect the recharge of soil water (Robinson et al., 2006), hinders vegetation sustainability and may evolve towards soil degradation at a later stage (Ashton and Kelliher, 1996; Breshears et al., 2005). Therefore, soil water supplement from rainfall has become the key determinant for the maintenance and sustainability of existing vegetation.

Soil water content is highly dependent on water infiltration and groundwater. Under certain conditions, roots create preferential flow paths, and preferential flow can occur via root channels, which sometimes occupy up to 70% of soil macropores (Noguchi et al., 1999). Preferential flow means that water and solutes unevenly and rapidly move through soil of which macropores and cracks participate in most of the flow. When the soil receives replenishment on the entire inflow boundary, water and solutes bypass the soil matrix and move quickly through a small part of the soil. Alive and dead plant roots can both form the preferential flow paths that favour water infiltration, especially in dried areas (Newman et al., 2004; Bogner et al., 2010; Ghestem et al., 2011; Wu et al., 2021). Therefore, the macropores formed by root channels are one of the important processes of surface water infiltration. Recent studies have indicated that roots have significant effects on soil water infiltrability (Wu et al., 2016; Huang et al., 2017; Cui et al., 2019; Liu et al., 2019, 2020), and the traits of plant roots have important effects on the preferential flow (Jiang et al., 2018; Wu et al., 2021). Root system characteristics (such as root diameter, root density, etc.) determine the decay and decomposition rate of the root system. In particular, the decay of plant roots is related to the vegetation type and root density, because the root diameter and length determine the root channel development (Ghestem et al., 2011).

Alfalfa (*Medicago sativa* L.) is one of the most important forage crops and planted widely in the world because of its high nutritive value, drought-resistance and good adaptability to drought and poor soil conditions (Li et al., 2007). By studying the water infiltration rate of alfalfa fields, Wu et al. (2019) found that root decay would improve soil structure and promote rainfall infiltration. Meanwhile, shrubs are the main vegetation type in drylands (Cushman et al., 2010), and Korshinsk peashrub (*Caragana korshinskii* Kom.) is widely cultivated in sandy grasslands and arid areas of northern China and Mongolia (Deng et al., 2017) because it is a long-lived legume shrub with good drought tolerance. However, the implementation of large-scale planting programs of Korshinsk peashrub have triggered soil water deficit and caused dried

soil layers (Gardiol et al., 2003; Maestre et al., 2009). Red plum apricot (*Armeniaca vulgaris* Lam.) was planted in large areas in arid Chinese regions to improve farmers' income. However, the occurrence of DSL caused by vegetation restoration, especially due to the limitation of rainfall supplement, has caused an increase in the vegetation mortality, threatening the sustainable development of agricultural ecosystems in water-scarce regions.

Because of the planted vegetation death, plant roots have decayed and formed large channels, favouring rainfall infiltration. This fact have created favourable conditions to improve soil water status when rainfall happens, providing a better growing environment for the subsequent plants' growth and maintenance (Wu et al., 2019). A number of studies have demonstrated that gramineous plants with a thin and shallow root system, and leguminous plants with taproot and deep root systems, affect soil water infiltrability differently due to their different root architectures (Huang et al., 2017; Zhang et al., 2019). Especially different plant types, such as grass, shrubs and trees have different effects on soil infiltrability. Although the soil water consumption of trees and shrubs is higher than that of grasses (Shangguan, 2007), the effects of decayed root channels on soil water storage are still not clear (Wu et al., 2016; Jiang et al., 2018; Guo et al., 2019). Root decay favours restoring soil water storage, especially by promoting the supplement of water into the deep soil layers (Germann et al., 2012; Guo et al., 2019; Wu et al., 2021). Thus, the replenishment of deep soil moisture by decayed root systems may be an effective way to mitigate dried soil layers with a minimum environmental impact, and this process could provide better soil water conditions for future plants.

This study examines the contribution of preferential water flow through decayed root channels on soil water replenishment in a semi-arid Chinese region, focuses on three vegetation types, namely: Korshinsk peashrub, red plum apricot and alfalfa (data from Guo et al., 2019). In particular, two goals are foreseen to: (1) examine the effects of the decayed roots on the soil water infiltration rates, and (2) determine the contribution of the preferential flow from the decayed roots to soil water replenishment by comparison with the observed flow in alive root plots. This study aims to better understand the effects of root channels of different degraded vegetation types on soil moisture and infiltrability, which can provide a theoretical basis for soil water supplement in arid and semi-arid regions.

2. Materials and methods

2.1. Study areas

The experimental site is located in Shanghuang village, Hechuan township (35°59'–36°02'N, 106°26'–106°30'E), Yuanzhou district, Guyuan city, in Ningxia Hui Autonomous Region at an altitude of between 1530 and 1822 m a.s.l. This study area is located in the western part of the Loess Plateau, China, and has a semi-arid cold climate. The average annual precipitation is about 419 mm, but it is widely variable and about 70% of the total precipitation occurs between May and October. The average annual temperature is 6.9 °C. The soil in the study area is classified as Luvisol (WRB, 2014) with 26.6% clay, 62.0% silt and 11.4% sand (Chai et al., 2019). The soil pH value is about 8.2–8.4 and the organic carbon content in the topsoil is about 5–8 g/kg. Since 1980s, a native Korshinsk peashrub has been used to restore the degraded sloping lands in most areas (ca. 90%). In addition, large areas are cultivated with red plum apricot, which is the main economic crop in the area.

2.2. Experimental design

We selected three contrasted vegetation types that are among the major crops and afforestation species in the study area: Korshinsk peashrub (*Caragana korshinskii* Kom.), red plum apricot (*Armeniaca vulgaris* Lam.) and alfalfa (*Medicago sativa* L., the data from Guo et al., 2019). Bare land plots were established as control site (Fig. 1). Korshinsk peashrub is a dominant species in this area because it can easily survive in arid and semi-arid environments. Red plum apricot not only can easily survive in arid and semi-arid environments, but also has a great economic value for local farmers. Hence, red plum apricot orchards cover large areas in this region. In order to obtain comparable values between the new plots (data of the Korshinsk peashrub and fruit tree plots) and the data of our previous study (alfalfa grassland, see Guo et al., 2019), we replicated the treatment conditions of the previous study as follows: Selection of alive and decayed specimens of Korshinsk peashrub and red plum apricot. The bare land coincides with the space in between the three vegetated plots in such a way that the control area has similar soil properties to each treatment. The same experimental procedure was performed in both experiments.

Thirty years ago, Korshinsk peashrub was planted in this area and nowadays it has become the dominant species. However, with the increase of the planting age, the older stands have been degraded. In order to eliminate the influence of human trampling on the soil properties during fruit harvesting, we selected an abandoned fruit tree plantation for experimentation. There are no grazing animals in the red plum apricot. Therefore, the experimental site does not have any land management procedures that affect soil compaction. The plantation of red plum apricot started 40 years ago, but some plants died about 5 years ago due to severe water shortage. The information about the planting time and death time of the two vegetation types (Korshinsk peashrub and red plum apricot) was obtained by data consulting and interviews with local residents.

All treatments included four repeats. To ensure data reliability, the infiltration data were analyzed after infiltration experiments. When infiltration data included abnormal values within the same treatment, the number of repetitions increased. Hence, there is an uneven number of

repetitions of treatments in this study. The following experimental design of field measurements was carried out: Bare land (4 repetitions), decayed Korshinsk peashrub (7 repetitions), alive Korshinsk peashrub (5 repetitions), decayed red plum apricot (8 repetitions) and alive red plum apricot (6 repetitions). The selection of these experimental points was completely random.

2.3. Measurement of infiltration rates

Water infiltration processes were evaluated using a double-ring infiltrometer. The operation steps of the entire infiltration process are based on the protocol of Zhang et al. (2017). The double-ring infiltrometer is composed of a 16-cm diameter inner ring and a 32-cm diameter outer ring. It was made using 1-cm wall thickness and 20-cm height PVC pipes. The purpose of the outer ring is to have the infiltrating water act as a buffer zone against internal water movement. Just before the infiltration test, the litter and plants (red plum apricot, Korshinsk peashrub and alfalfa) were removed from the soil surface before inserting the infiltrometer. For reducing the influence of artificial disturbance on soil structure, the infiltrometers were gently and vertically inserted into the soil about 5 cm (Fig. 1). The insertion position ensured that the root collar was located in the center of the infiltrometer. Then, water was added rapidly and simultaneously to the inner (with methylene blue) and outer (without methylene blue) rings at 5 cm depth. Water level of the inner ring and the outer ring was maintained consistently. Methylene blue is an important tool to visualize preferential flow pathways and areas within the soil profile as flow paths. The time was recorded each time the water level in the inner ring dropped 1 cm, and measurements stopped when they reached three roughly consecutive equal values. Water was refilled up to the 5 cm height when the water line dropped to 1 cm over the course of the experiment. When the infiltration time did not change after three consecutive measurements, the corresponding rate was assumed to be the steady-state infiltration rate (SIR). The initial infiltration rate (IIR) was calculated as the mean value of the first 3 min. Similarly, the steady-state infiltration rate was calculated approximately using the last three values. The infiltration rates represented the matrix infiltration and that through the alive or decayed roots.

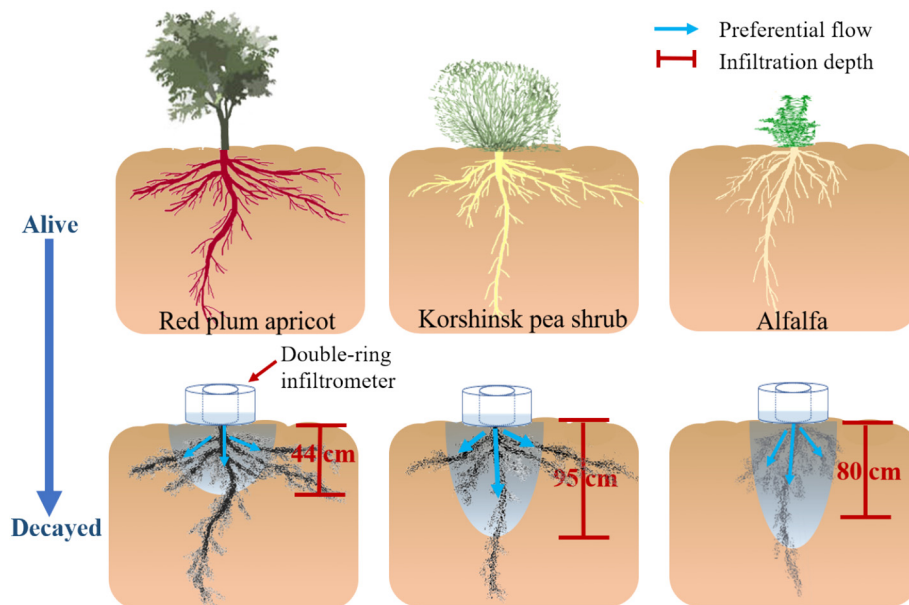


Fig. 1. Experimental diagram of the red plum apricot (*Armeniaca vulgaris* Lam.), Korshinsk peashrub (*Caragana korshinskii* Kom.) and alfalfa grassland (*Medicago sativa* L.; data source: Guo et al., 2019). This figure shows the alive and decayed roots and the average depth of water infiltration associated with the decayed roots. The light blue area indicates the wetting front when measure the infiltration rate by using the double-ring infiltrometer. The red numbers represent the average depth of water infiltration under the same treatment. The infiltration depth was measured after the experiment stopped 24 h when the infiltration rate was stable.

2.4. Soil sampling and analysis

In order to drain the infiltration water, the vertical soil profiles were excavated along the perimeter of the double ring when the experiment ended. The vertical soil profiles of the red plum apricot, Korshinsk peashrub and alfalfa grassland plots are shown in Fig. 1. The wetted vertical profiles were recorded according to the wetted area. Soil bulk density (BD , g cm^{-3}) was obtained by means of using soil bulk samplers (100 cm^3), at a sampling interval of 10 cm up to 50 cm soil depth. Three replicates were taken at each soil layer. Oven-drying method was used to measure soil gravimetric water content (SWC , %). Soil gravimetric water content was measured by using a soil drilling method before infiltration measurement, and other parameters were measured after infiltration measurement. Total porosity (TP , %) of the soil layers was computed as:

$$TP = \left(1 - \frac{BD}{ds}\right) \times 100 \quad (1)$$

where ds is the soil particle density, and in this study it was assumed to be 2.65 g cm^{-3} (Huang et al., 2017; Schjønning et al., 2017).

2.5. Measurement of roots

The average root diameter (ARD , cm) in the inner ring was calculated from the root diameters (di , cm) measured with a Vernier caliper (precision of 0.1 mm) (Wu et al., 2017; Guo et al., 2019). Moreover, the number of roots (n) in the inner ring was also determined before the experiment to calculate the root area (RA , cm^2). ARD and RA were calculated as follows (Wu et al., 2017):

$$ARD = \sum_{i=1}^n \frac{di}{n} \quad (n = 1, 2, 3, \dots) \quad (2)$$

$$RA = n \cdot \pi \cdot \frac{ARD^2}{4} \quad (n = 1, 2, 3, \dots) \quad (3)$$

We used a digital camera to record the vertical profile to determine the extent of the wetted area after the experiment. The resolution and accuracy of the digital camera are 3968×2976 and 0.25 mm, respectively. Meanwhile, the lateral width and vertical depth of the wetted volume were measured with a 1000 mm steel ruler with precision of 1 mm.

Table 1

Mean $\pm$ SD values of soil water content (SWC), bulk density (BD) and total porosity (TP) of the four treatments at the soil depth of 0–50 cm, at 10-cm intervals.

Treatments	Soil depth (cm)	SWC (%)	BD (g cm^{-3})	TP (%)
Bare land	0–10	$15.83 \pm 1.10\text{ab}$	$1.25 \pm 0.05\text{a}$	$52.74 \pm 1.98\text{b}$
	10–20	$11.62 \pm 0.21\text{b}$	$1.26 \pm 0.02\text{a}$	$52.56 \pm 0.74\text{b}$
	20–30	$11.10 \pm 1.07\text{b}$	$1.22 \pm 0.03\text{a}$	$53.93 \pm 1.05\text{b}$
	30–40	$9.73 \pm 0.46\text{b}$	$1.17 \pm 0.05\text{a}$	$55.81 \pm 1.88\text{a}$
	40–50	$10.56 \pm 0.56\text{a}$	$1.13 \pm 0.01\text{a}$	$57.25 \pm 0.40\text{a}$
Korshinsk peashrub land	0–10	$16.15 \pm 0.14\text{a}$	$1.09 \pm 0.06\text{b}$	$58.93 \pm 2.22\text{a}$
	10–20	$14.43 \pm 1.14\text{a}$	$1.10 \pm 0.07\text{b}$	$58.42 \pm 2.64\text{a}$
	20–30	$14.84 \pm 0.56\text{a}$	$1.16 \pm 0.04\text{ab}$	$56.16 \pm 1.69\text{a}$
	30–40	$11.17 \pm 0.40\text{a}$	$1.12 \pm 0.05\text{a}$	$57.56 \pm 1.99\text{a}$
	40–50	$9.21 \pm 0.79\text{b}$	$1.17 \pm 0.03\text{a}$	$55.84 \pm 0.94\text{a}$
Red plum apricot land	0–10	$13.36 \pm 1.44\text{b}$	$1.15 \pm 0.06\text{ab}$	$56.50 \pm 2.28\text{ab}$
	10–20	$12.69 \pm 1.60\text{ab}$	$1.08 \pm 0.06\text{b}$	$59.17 \pm 2.42\text{a}$
	20–30	$9.43 \pm 2.70\text{b}$	$1.19 \pm 0.02\text{a}$	$55.04 \pm 0.65\text{b}$
	30–40	$6.78 \pm 0.37\text{c}$	$1.14 \pm 0.07\text{a}$	$56.86 \pm 2.78\text{a}$
	40–50	$7.23 \pm 0.17\text{c}$	$1.21 \pm 0.06\text{a}$	$54.40 \pm 2.30\text{a}$
Alfalfa grassland	0–10	$2.67 \pm 0.02\text{a}$	$1.31 \pm 0.03\text{a}$	$50.10 \pm 1.24\text{a}$
	10–20	$3.87 \pm 0.86\text{a}$	$1.33 \pm 0.58\text{a}$	$49.98 \pm 1.94\text{a}$
	20–30	$3.69 \pm 0.53\text{a}$	$1.48 \pm 0.05\text{a}$	$44.15 \pm 2.04\text{a}$
	30–40	$4.19 \pm 1.26\text{a}$	$1.38 \pm 0.07\text{a}$	$47.75 \pm 2.69\text{a}$
	40–50	$4.07 \pm 0.92\text{a}$	$1.23 \pm 0.03\text{b}$	$53.42 \pm 1.28\text{a}$

Note: For each soil depth but different treatment, the values followed by a different letter are significantly different at 0.05 level. Data of soil water content, soil bulk density and total porosity of alfalfa were cited from our previous study reported in Guo et al. (2019).

2.6. Statistical analysis

The one-way analysis of variance (ANOVA) and the least significant difference (LSD) test were applied to examine the differences of the soil characteristics in the same soil layer within the different experimental treatments. Significant difference level was set at 0.05. The correlation between ARD , RA and the steady infiltration rate was evaluated by using regression analysis. All statistical analyses were carried out using SPSS 22.0 (IBM, USA) software. All figures were created using SigmaPlot 14.0.

3. Results

3.1. Soil physical characteristics in the different treatments

The highest SWC appeared in the topsoil layer (0–10 cm) in all treatments with the highest values in the Korshinsk peashrub land ($\overline{SWC} = 16.15\%$), followed by the bare land ($\overline{SWC} = 15.83\%$) and the red plum apricot land ($\overline{SWC} = 13.36\%$), whereas the lowest SWC was found in the alfalfa grassland ($\overline{SWC} = 2.67\%$) (Table 1). The differences of SWC between the treatments were significant at all soil layers. In particular, in the 0–40 cm soil layer, the SWC of the Korshinsk peashrub plots was higher than in the other experimental treatments. In the bare soil plots, SWC was significantly higher than in the other treatments at the 40–50 cm soil layer. An overall general trend was observed, showing lower SWC with increasing soil depth in all experimental treatments. Regarding BD , the measured values decreased with increasing soil depth in the bare land, whereas increasing BD values were found in the other treatments. The BD of the bare land in the 0–30 cm soil layer was significantly higher ($\overline{BD} = 1.24 \text{ g cm}^{-3}$) than in the Korshinsk peashrub land and red plum apricot land ($\overline{BD}$ between 1.12 and 1.14 g cm^{-3}).

3.2. Water infiltration rates increased due to the presence of decayed plant roots

In general, the initial infiltration rate (IIR) of the bare land was the lowest (279.07 mm h^{-1}) among all treatments (Fig. 2). The IIR of the Korshinsk pea land and red plum apricot land with alive and decayed roots increased significantly compared to that of the bare land, and increased approximately 37%, 57%, 58% and 68%, respectively (Fig. 3).

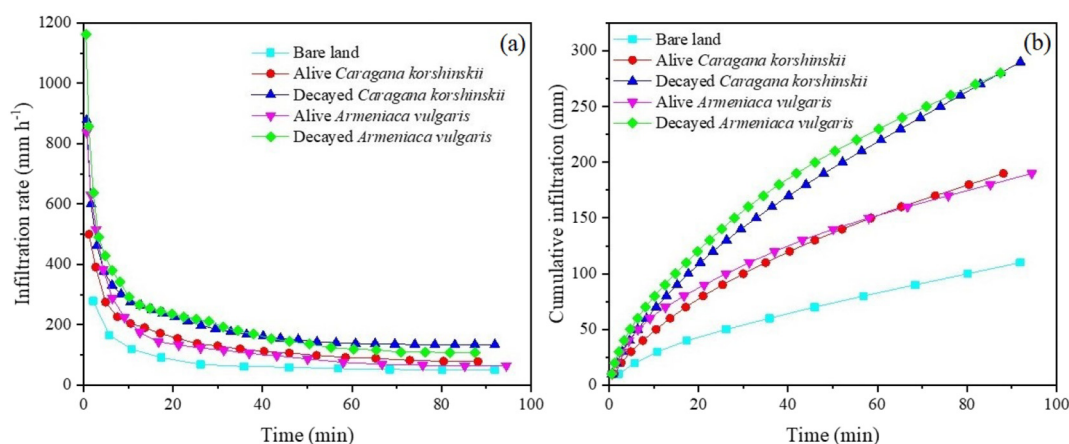


Fig. 2. Infiltration rate (a) and cumulative infiltration (b) curves of the bare land, and Korshinsk peashrub (*Caragana korshinskii* Kom.) and red plum apricot (*Armeniaca vulgaris* Lam.) plots.

The IIR of the red plum apricot plots with alive roots increased by 48% compared with that of Korshinsk peashrub plots with alive roots. However, the IIR of Korshinsk peashrub plots with decayed roots increased by 37% compared to the IIR of the red plum apricot plots with decayed roots. In addition, the steady infiltration rates (SIR) of the Korshinsk peashrub and red plum apricot plots with decayed roots were about 68% higher than the SIR of the plots with alive roots of the same species. The SIR of the Korshinsk peashrub and red plum apricot plots with decayed roots advanced 83% and 48% compared with the SIR of the bare land, respectively (Fig. 3).

3.3. Correlation between decayed root channel properties and soil water infiltration rates

The average root diameter (ARD) and root area (RA) were both positively and significantly correlated to the steady infiltration rates (SIR) in the Korshinsk peashrub and red plum apricot plots with both alive and decayed roots (Fig. 4). It is worth noting that the average SIR of the decayed root plots increased by 40.88% and 14.21% in the Korshinsk

peashrub and red plum apricot plots compared with the same species plots with alive roots, when the ARD was similar (Fig. 4a, c). Under similar RA values, the average SIR from the decayed roots increased by 41.09% and 14.69% in the Korshinsk peashrub and red plum apricot plots compared with the alive root plots (Fig. 4b, d). Furthermore, the infiltration depth in the Korshinsk peashrub plots with decayed roots was significantly higher than that in the red plum apricot plots with decayed roots (Fig. 5).

4. Discussion

Our results showed that soil water content was higher in the topsoil layer and decreased with increasing soil depth. This may be due to rainfall was the only source for soil water supplement, while the small amount of rainfall and compact soil make the rainfall confined in the shallow soil layer. It is known that root systems play an important role in forming macropores and influencing soil water infiltration (Lesturgez et al., 2004; Cui et al., 2019). Thus, the effects of plant root characteristics on soil hydrological processes play an important role on vegetation sustainability in drylands (Costantini et al., 2015; Neris et al., 2012). The macropores formed by the decayed roots act as important soil water supply sources for the deep soil layers (Bogner et al., 2010; Wu et al., 2019, 2021). This study shows that the soil infiltration rate of the Korshinsk peashrub and red plum apricot plots significantly increased the soil infiltration rate compared to the bare land.

Previous studies have shown that soil water infiltration rate could usually be affected by root characteristics, soil water content, and soil porosity (Wu et al., 2016; Cui et al., 2019; Guo et al., 2019; Zhai et al., 2019). By analyzing the differences in infiltration capacity of different artificial grasslands, Wu et al. (2019) found that soil water content, root biomass, and soil porosity were the main factors affecting soil water infiltration rate. This study found that the initial infiltration rate of the Korshinsk peashrub and red plum apricot plots with alive roots increased approximately by 37% and 57% compared to the initial infiltration rate of the bare land. The initial infiltration rate of the Korshinsk peashrub and red plum apricot plots with decayed roots significantly increased by 58% and 68% compared to the initial infiltration rate of the bare land. Studies have shown that soil water content was the main factor affecting the initial infiltration rate (Cui et al., 2019; Liu et al., 2019). The matric suction decreases with the increase of soil water content, which affects the rate of soil water infiltration (Rahardjo et al., 2018). The growth of vegetation roots can absorb water in the soil, thereby reducing the water content of the soil and increasing the suction of the substrate (Rahardjo et al., 2018). Although the Korshinsk peashrub had higher soil water content than bare land, their higher soil porosity could increase the rate of soil water infiltration. Cui et al. (2019) also

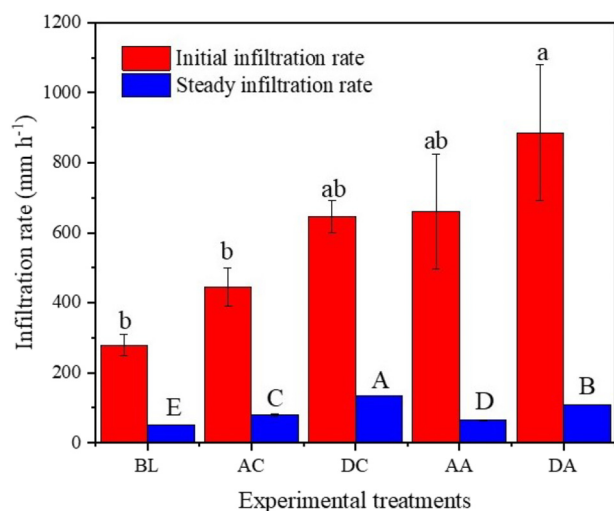


Fig. 3. Mean $\pm$ SD values of the initial infiltration rate and steady-state infiltration rate of the different experimental treatments. Different letters between experimental treatments indicate significant differences at 0.05 level. Note: BL: bare land; AC: alive Korshinsk peashrub land (*Caragana korshinskii* Kom.); DC: decayed Korshinsk peashrub land (*Caragana korshinskii* Kom.); AA: alive red plum apricot land (*Armeniaca vulgaris* Lam.); DA: decayed red plum apricot land (*Armeniaca vulgaris* Lam.).

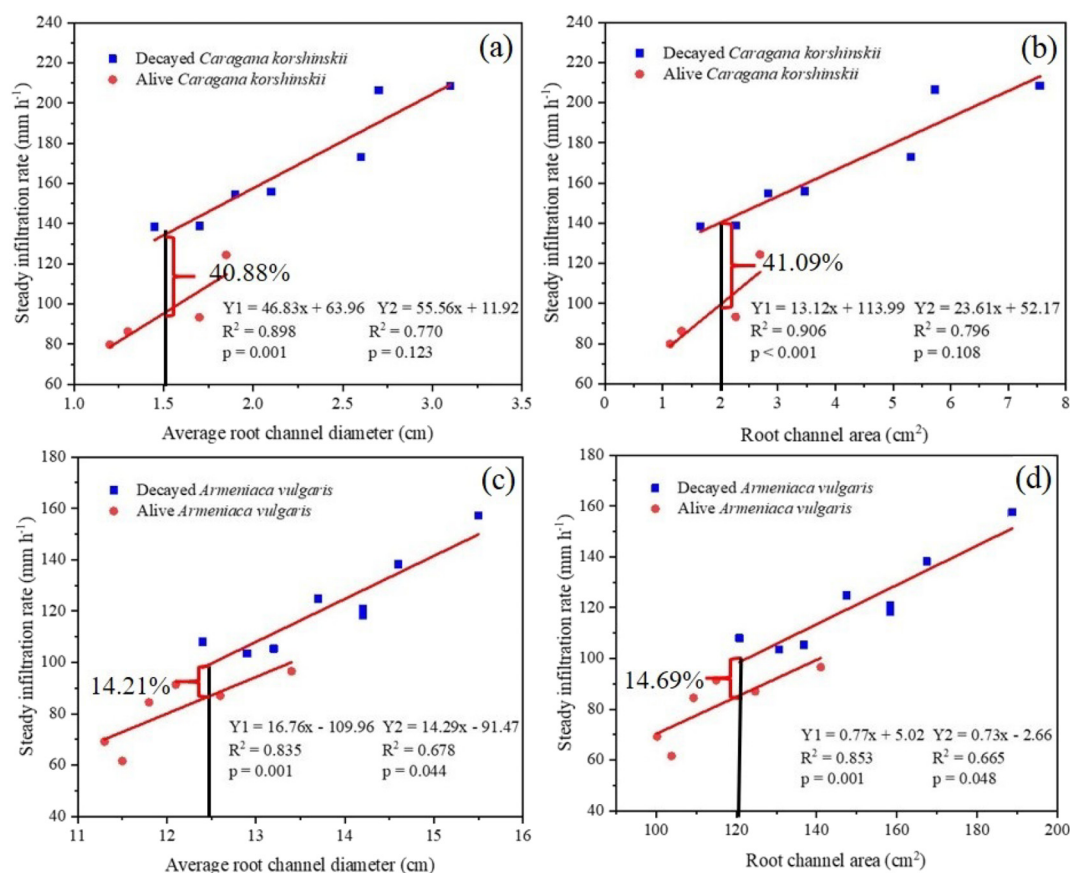


Fig. 4. Correlation between the average root channel diameter and the steady-state infiltration rate of the plots with decayed and alive roots of Korshinsk peashrub (*Caragana korshinskii* Kom.; A) and red plum apricot (*Armeniaca vulgaris* Lam.; C). Correlation between the root channel area and the steady-state infiltration rate of the plots with decayed and alive roots of Korshinsk peashrub (*Caragana korshinskii* Kom.; B) and red plum apricot (*Armeniaca vulgaris* Lam.; D). Note: Y1 for the decayed root, Y2 for the alive root. The values 40.88%, 41.09%, 14.21%, and 14.69% indicate the average increased percentage of the steady-state infiltration rate by preferential water flow through decayed plant roots when the diameter of the roots was similar.

found that the soil porosity has a positive and significant relationship with the initial infiltration rate.

The soils in the initial infiltration stage were unsaturated soils, so the infiltration rate at this stage reflected the downward flow rate of water in the soil under unsaturated conditions (Rahardjo et al., 2018; Zhai et al., 2019). The water infiltration rate under the unsaturated and saturated conditions of the soil is an important index to evaluate the water infiltration capacity in the soil during rainfall (Rahardjo et al., 2018; Zhai et al., 2019). In this study, soil moisture gradually became saturated with the increase of infiltration time, so the infiltration rate in the stable infiltration stage could be approximated as the saturated infiltration rate of the soil. This study showed that the steady-state infiltration rates of both the alive and decayed roots increased compared to the bare land, and more important, this pattern was pronounced in the plots with decayed roots. In general, both the alive and decayed roots could form relatively large, continuous, well-connected and open channel networks, considering preferential flow paths (Aubertin, 1971). Roots can release large amounts of exudates that improves soil structure, and further influences soil infiltrability (Bronick and Lal, 2005; Wu et al., 2019). Gerke et al. (2015) reported that biomat flows can transmit considerable lateral subsurface flow in some hillslope soils. Moreover, soil preferential flow caused by root channels has crucial effects on water infiltration process, which is conducive to the recharge of groundwater and has great significance in predicting runoff generation (Weiler and Naef, 2003). Our results show that the preferential flow through the root channels in the Korshinsk peashrub plots and fruit tree plantation with decayed roots significantly improved soil water infiltration compared with the measured infiltration rates in the plots with alive roots and those of bare land. Besides, the SIR of the

Korshinsk peashrub and red plum apricot plots with decayed roots improved by 83% and 48% compared with those observed in the bare land.

Our previous studies reported that the decayed roots in alfalfa grassland increased the SIR by 31.8% compared to the bare land (Guo et al., 2019). Compared with the alive roots, the decayed roots were more likely to form long, continuous and stable paths (Mitchell et al., 1995). The decayed plant roots provided new space for water because of a clear porous soil with lower density and higher porosity. Combined with our previous research findings (Guo et al., 2019), there were significant differences in infiltration rates between the dead and living roots. This result may be explained by the coarser roots of the fruit trees than those of the shrub and grasses, which led to greater decay rates of the root system in the shrubs and grasses that could allow a favourable flow of water. Compared to shrubs, grasses have finer roots that decay more rapidly, and thus, the subsequent root channels are finer. This fact can explain why the infiltration variation rate was lower in the grasses than in the shrubs. We also found that the decayed roots significantly improved the infiltration capacity of soil water in a more efficient way than the alive roots under the same root diameter conditions. Our results provide clear evidence that the decayed root systems of the three vegetation types act as preferential flow channels for increasing soil water supplement under semi-arid conditions. The significant and positive effects of root channels on preferential flow was conducive to the replenishment of the soil water in deep layers, and thus, to mitigate dried soil layers and promote existing vegetation sustainability in arid and semi-arid regions (Ngao et al., 2007). Vegetation usually accelerates the consumption of soil moisture in the early stage of establishment, which leads to soil desiccation (Maestre et al., 2009; Jia et al., 2017; Huang et al., 2018). However, the above-ground parts

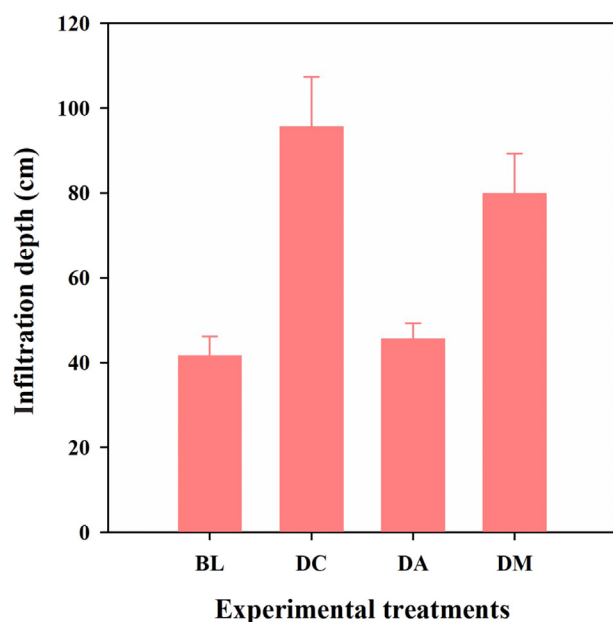


Fig. 5. The soil water infiltration depth of the different experimental treatments (mean $\pm$ SD). Note: The infiltration depth is the maximum depth that the water reached along the soil profile after 24 h. Different letters between experimental treatments indicate significant differences at 0.05 level. BL: bare land; DC: decayed Korshinsk peashrub land (*Caragana korshinskii* Kom.); DA: decayed red plum apricot land (*Armeniaca vulgaris* Lam.); DM: decayed alfalfa grassland (*Medicago sativa* L.; data source: Guo et al., 2019).

and roots of vegetation gradually die and decay with the continuous decrease of soil moisture (Jia et al., 2017; Huang et al., 2018; Wu et al., 2021), which increases the rate of soil water infiltration and replenishes soil moisture. The decayed root channel of dead plants may become an important way of deep soil water replenishment. This may be a synergistic process of interaction and adaptation between vegetation and soil moisture. Thus, this solution to alleviate the dried soil layers caused by inappropriate afforestation may become a natural compensation mechanism for dry soil.

The distribution of roots in the soil profile may lead to clear differences in infiltration rates and infiltration depth. Our results indicate that the average infiltration depth of the shrub plots was significantly higher than that of the fruit tree plantations (Fig. 5), which may result from the undecayed root system of the orchard. Normally, the root system of fruit trees is deeper than that of the shrub and grasslands. However, the early death of the fruit trees in our study area (just five years old) as well as the lower decay rate may induce the lower water infiltration by preferential flow from decayed tree roots. Thus, the effects of the decayed degree of plant roots on preferential flow and soil water recharge should be considered in future studies. This task is of particular importance in water-scarce areas, and further studies should be focused on the preferential flow related to complex root morphologies and different decay degrees in different vegetation types.

5. Conclusions

The effects of decayed root channels on soil water infiltrability were demonstrated in distinct vegetation types adapted to semi-arid areas, showing clear differences among those channels formed by the decayed and alive roots. Combining the author's previous research findings, we concluded that shrub species had the highest value of steady infiltration rates, followed by fruit tree and grassland species. Our findings clearly proved that decayed root channels improve soil water infiltration rates compared with the rates obtained in the alive root plots of the same plant species. Under the same root diameter conditions, the

steady infiltration rates of vegetation with decayed roots were significantly higher than the infiltration rates with alive roots; altogether, the decayed roots significantly increased the soil infiltrability compared to the alive roots. The channels generated after the death of the plants may become an important natural compensation mechanism to dried soil layers caused by inadequate afforestation activities. Our findings contribute to better understanding the role played by the decayed root channels of different degraded vegetation types on soil water supplement in semi-arid regions. This process of soil water replenishment is environmentally friendly and matches the target of land degradation neutrality.

CRedit authorship contribution statement

Zeng Cui: Investigation, Data curation, Writing – original draft, Formal analysis. **Ze Huang:** Investigation, Data curation, Writing – original draft, Formal analysis, Visualization. **Yu Liu:** Investigation, Data curation, Project administration, Supervision. **Manuel López-Vicente:** Writing – original draft, review & editing. **Gao-Lin Wu:** Conceptualization, Methodology, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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